

Animesh Saha

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EDUCATION

University at Buffalo, The State University of New York

Bachelor of Science in Computer Science

GPA: 3.86

Buffalo, New York

Expected, Dec 2025

SKILLS

Languages: Java, Python, C, OCaml, PHP, JavaScript, HTML/CSS

Frameworks/Libraries: TensorFlow, OpenCV, Pillow, Pandas, Selenium, React.js, JQuery, Node.js, NumPy, Matplotlib

Tools: Git, JUnit, Streamlit

PROJECTS

Song Recommender | *React, CSS, MySQL, PHP*

October 2024

- Built a full-stack music recommendation application with REST API, using React, PHP, MySQL
- Integrated several Spotify API to predict the most probable recommendation for the users and developed OAuth for user authentication
- Implemented core user functionalities, including account login, password reset, and playlist creation, with an emphasis on agile development practices and system design concepts

Draw it Right | *Python, OpenCV, Mediapipe, TensorFlow, Streamlit*

Winner, UB Hacking 2024

November 2024

- Created a finger-tracking tool using OpenCV and Mediapipe, allowing gesture-based virtual drawing by detecting specific finger movements
- Integrated a model to detect and classify user-drawn objects in real time with an accuracy of 92.5%
- Leveraged Streamlit to create an interactive and user-friendly interface for seamless functionality

OCaml Interpreter | *OCaml*

February 2025

- Built a full-fledged interpreter from scratch in OCaml, following the concepts of functional programming, closures, scoping
- Implemented core language features including lexical scoping, higher-order functions

Facial and Hand Written Digits Recognition | *Python, Pandas, Numpy, Matplotlib*

March 2025

- Built and trained a neural network model from scratch to classify hand-written digits (MNIST) and facial structures with accuracy of 91% of the testing data
- Implemented a Convolutional Neural Network with ReLU activation that resulted in improved accuracy of 94.6% of the testing data
- Improved testing accuracy to 96.8% and mitigated overfitting by fine-tuning hyperparameter, L2 regularization, dimensionality reduction with Kernel PCA

EXPERIENCE

Teaching Assistant

August 2025 – Present

CSE 305: Programming Language

Department of Computer Science and Engineering, University at Buffalo

- Conducted Office hours to help more than 80 students debug complex programming projects and teach functional programming concepts
- Provided solutions to specific questions asked by more than 65 students online and in person and graded assignments and exams

Academic Tutor

August 2024 – Present

Tutoring and Academic Support Services

University at Buffalo

- Debugged and helped students test complex programming problems and demonstrated visualization for algorithms such as BFS, DFS, Djisktra, A* to build deep algorithmic understanding
- Tutored more than 60 students and organized one-on-one sessions with students to explain data structures and algorithm as well as Calculus 2
- Conducted drop-in tutoring to provide tutoring to more than 600 students as a campus living tutor

Student Assistant

August 2023 – Present

Special Programs Management, School of Engineering and Applied Science

University at Buffalo

- Led a team of 5 to coordinate the Dean's Advisory Council meetings, ensuring all organizational goals were met
- Reviewed over 50 resumes and conducted interviews to select the most qualified candidates for various positions

AWARDS AND HONORS

Tau Beta Pi: Spring 2025 Inductee (Nation's oldest Engineering Honor Society, awarded to top 12.5% of class based on academic excellence)

Dean's List: Spring 2025 — **Dean's List:** Spring 2024 — **Dean's List:** Fall 2023